CS4.406 Assignment 1 — Lexical & Semantic Retrieval: Design Note

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1 What I built

Raw archives → per-dataset readers → **one unified schema** → temporal split → parquet feature store (42 s rebuild, both datasets). Two retrievers behind one `Retriever` protocol — BM25 over title+abstract, and MiniLM embeddings with a brute-force exact index — plus recency, popularity, random, RRF fusion and a popularity-prior blend. One `evaluate()` scores all of them, so Q4.5 holds by construction: there is nowhere else a metric can be computed.

Two evaluation regimes are kept strictly apart, because conflating them produces numbers that look fine and mean nothing. *Corpus regime*: recall@K over all 21K–65K articles — *did the clicked article survive the cut? Slate regime*: AUC/MRR/nDCG over the ∼11–40 candidates actually shown — *did you rank it above the others?* Every result row carries a `regime` column.

2 The constraint that drove every design choice

Every richer feature EB-NeRD offers is one MIND lacks: body text, click timestamps, publish time, session ids (0% on MIND), provided embeddings. Using any of them for a headline number silently converts a *dataset* comparison into a *feature-availability* comparison.

Rule adopted: the shared path uses only the intersection — title, abstract, category, click history. Everything one-sided becomes a labelled single-dataset ablation. This one rule resolves the encoder choice, recency weighting, body text and session context together.

3 Key decisions and rejected alternatives

Decision	Chosen	Rejected, and why
Corpus	Union of MIND's two `news.tsv` (65,238)	Per-split: 23.3% of dev clicks are on train-absent articles, an **invisible 0.767 recall ceiling** [F2,F20]
Split	Hold out official test, carve val from train tail	Literal "7 days test": MIND's *entire* labelled span is 6.0 days [F27]
Lexical fields	Title + abstract (Q2.1 mandates)	Title+body — MIND has no body, breaking comparability [F10]
Encoder	**MiniLM 384-d**, mean-pooled, L2-normalised	**XLM-R fails the Danish probe** (below); `[CLS]` pooling; per-language stemming
Index	Brute force (exact; Q3.2 permits it)	HNSW — both corpora are <200K, where exact costs 190–890 ms and is the recall *ceiling* ANN is measured against
User vector	Log positional decay, n=20, conditional pooling	Exponential decay collapses the query to 2–3 clicks; plain mean blurs multi-interest users
Session context	**Dropped**, deferred to C-2	0% coverage on MIND; EB-NeRD sessions average **1.9 impressions** [F29]
Coverage CI	Point estimate, marked `(no CI)`	Every bootstrap scheme is biased low — see §4.6

4 Observations

4.1 Recency dominates; text matching is secondary [F16,F21]

On EB-NeRD, **94.3% of clicks are on articles under 24 h old** while only ∼1.1% of the corpus is that fresh. A retriever that *ignores the user entirely* and returns the newest articles scores **recall@50 = 0.8986**, against BM25's 0.0043 over the full corpus. Restricting either retriever to a 24 h window moves it from ∼0.004 to ∼0.24 — a **33× swing from the candidate pool, not the scoring function**.

Candidate generation on news is primarily a *freshness* problem. This is legitimate, not a leak: publish time is known at serving time; ranking by *future* engagement would not be.

4.2 Lexical vs. semantic: the answer flips between datasets [Q3.5]

recall@100, n=4,000	**EB-NeRD (Danish)**	**MIND (English)**
Best overall	recency **0.9434** [0.9361, 0.9505]	popularity **0.0688** [0.0615, 0.0763]
Lexical	bm25+24h **0.2375** [0.2244, 0.2490]	bm25 0.0142 [0.0108, 0.0172]
Semantic	sem+24h 0.2307 [0.2175, 0.2432]	sem **0.0163** [0.0130, 0.0199]
RRF fusion	0.0070 — *no gain*	**0.0181** [0.0146, 0.0218] — *best retriever*

The point estimates favour semantic on MIND and lexical on EB-NeRD — but a paired bootstrap on the difference finds *neither gap significant*: MIND fusion vs BM25 nDCG@10 +0.0033 [−0.0029, +0.0105] (they differ on 55 of 800 impressions); EB-NeRD semantic vs BM25 −0.0055 [−0.0277, +0.0155] (differing on 470 of 800 — ample signal, no detectable gap). At n = 4,000 the two families are indistinguishable and the apparent flip is not established.

The leaderboard disagrees, and that is the finding. Submitting fusion for MIND scored **AUC 0.5934 against BM25's 0.5568** — a real +0.0366 on 2.37M impressions, an effect the offline harness had no power to see. An offline proxy this size can rule out large differences; it cannot rank retrievers.

The slice tells a sharper story. EB-NeRD recall@100, cold defined as history ≤ 97 (q0.25 of observed; median 235):

	cold (n=1,004)	**warm** (n=2,996)
semantic + 24h	**0.2468** [0.2198, 0.2742]	0.2253 [0.2104, 0.2407]
bm25 + 24h	0.2216 [0.1982, 0.2490]	**0.2428** [0.2276, 0.2588]

The ordering crosses over: semantic wins cold users, lexical wins warm ones. With few clicks, word overlap has too few terms to work with while embeddings still place a sparse history somewhere meaningful. CIs overlap, so this is suggestive, not established — but the crossover is the shape, and it is what Q3.5 asks for.

4.3 XLM-RoBERTa's collapse is anisotropy — and it is fixable [F37,F71]

The brief names BERT/XLM-RoBERTa. Before trusting either, a probe embedded 5 obviously-related Danish headline pairs and 5 obviously-unrelated ones:

Encoder	related	unrelated	margin	verdict
`xlm-roberta-base` (768-d)	0.9972	0.9954	**+0.0018**	**OVERLAPS**
MiniLM (384-d)	0.6523	0.0253	**+0.6271**	SEPARATES

XLM-R rates *"Brøndby beat FCK"* and *"a new apple cake recipe"* as 0.995 similar. It is trained for masked-language modelling, not to make cosine similarity meaningful, and its vectors collapse into a narrow cone. MiniLM achieves a 348× larger margin from *half* the dimensions, so it ships; XLM-R is reported as a measured failure rather than a silent substitution.

But "cannot separate" was too strong a conclusion, and finding the limit of our own claim is the more useful result [F71]. Subtracting the mean direction — one line — takes the margin from +0.0018 to +**0.3875**, a **215**× improvement; with 128-d truncation it **separates cleanly** at +0.5070. The information was always there, drowned by one dominant direction. Whitening, the textbook fix, *fails* (+0.0020) — it removes the signal with the anisotropy.

Most tellingly: even at the collapsed baseline, **4 of 5 related pairs still rank in the top 5 by cosine**. Retrieval consumes *ranking*, not absolute similarity, so the property the system depends on was largely intact while the property the probe measured was destroyed. **A smoke test can prove something unusable only if it tests the thing you use** — the same lesson as the BM25 `score_subset` check, where verifying rank agreement rather than absolute scores was the right call. MiniLM still wins on both axes (+0.6270 baseline, 5/5 on ranking), so the shipped choice is unchanged; what changes is the strength of the claim. *Caveat:* $n = 5$ *pairs is a smoke test, and no retrieval metric was run on centered vectors — no recall improvement is claimed.*

4.4 Slate metrics barely discriminate — always show the random row [F25]

EB-NeRD slates average 11–12 items with exactly one click, so **random ranking scores nDCG@10 = 0.4367 and AUC = 0.4933**. The best retriever reaches 0.4750 / 0.5407. Every AUC sits within noise of 0.5. Any slate-regime table without a random baseline row is misleading, and recall@K in the corpus regime is the only metric that clearly separates retrievers here.

4.5 Conditional pooling earns its complexity on one dataset only

Pooling strategy counts over 4,000 impressions: EB-NeRD `mean` 3,991 / `recent_half` 9; MIND `mean` 2,117 / `recent_half` 1,462 / `max_pool` 270. **46% of MIND users have interests too scattered for a centroid, against 0.2% on EB-NeRD** — consistent with MIND's shorter, more varied histories (median 19 vs 92 clicks).

4.6 One metric legitimately has no confidence interval

Q4.4 demands a bootstrap CI on every metric. **Coverage cannot honestly have one.** It counts *distinct* articles, so it grows with sample size; resampling n units with replacement makes ∼37% duplicates, so every resample sees fewer unique articles. The first attempt produced 0.9783 [0.9035, 0.9235] — **the point estimate outside its own interval**. Subsampling without replacement fails at every ratio from 0.5 to 0.99. Reported as a point estimate marked `(no CI)`: saying so beats shipping a plausible-looking number that is biased by construction.

4.7 Embedding width: 256-d beats 384-d, and 128-d costs nothing [F47]

Dimension is the one semantic knob with a direct resource cost — memory, index size and search time all scale linearly in it. Truncating the cached 384-d MiniLM vectors and re-normalising isolates it exactly: same model, same training, only the width changes.

Dim	Vectors	recall@50	Paired difference vs 384-d
384	31.9 MB	0.2325	—
256	**21.2 MB**	**0.2500**	+0.0175 [+0.0025, +0.0338] **SIGNIFICANT**
128	**10.6 MB**	0.2437	+0.0112 [−0.0112, +0.0338] — no worse
64	5.3 MB	0.2175	−0.0150 — degrades

256-d is significantly better than full width at 34% less memory, and 128-d is indistinguishable from it at a third. The tail dimensions of a non-Matryoshka embedding carry mostly noise for this task; dropping them removes variance without removing signal — the mechanism that makes PCA often improve retrieval rather than merely compress it. Combined with [F37], where XLM-R's 768-d lost to MiniLM's 384-d by 348× on the Danish probe: **the training objective and the noise floor decide quality, the width does not**.

4.8 Query length: an inverted U, and still not significant [F51]

Swept `last_n` *inside* the 24 h window, the regime we ship, rather than over the full corpus where everything scores under 0.008. recall@50: 0.2288 (5), 0.2313 (10), 0.2475 (15), **0.2531 (20)**, 0.2456 (30), 0.2338 (50) — a clear peak at 20. Paired against the shipped 15 it is +0.0056 [−0.0156, +0.0250], **not significant**, and neither is any other value. The shape matches the full-corpus sweep, so the window does not change this conclusion even though window and K do interact elsewhere. We ship 15 and report that 20 is a candidate, not an improvement.

4.9 Multi-query retrieval: built, measured, rejected [F48]

The stated weakness of a single user vector is that someone who reads football *and* recipes gets a centroid between the two, matching neither. Clustering the history and retrieving per centroid should fix that — and [F40] found **46% of MIND users** falling back from the mean, so the population exists.

Measured: **not significant on either dataset, at 19–36× the query cost** (MIND 57.5 s vs 3.0 s; paired +0.0009 [−0.0055, +0.0077]). The clustering fired properly — 1,518 of 1,564 MIND users received three centroids — yet the two variants differ on only **16 of 800** impressions.

So the blended centroid is not as useless as the theory assumes. It still ranks each interest's articles highly enough to reach a top-50, because *recall@K at K=50 is a forgiving target.* The blur would matter for precision at rank 1 — a re-ranker's problem, not candidate generation's. Keeping the single vector was the right call, and is now evidence rather than a guess.

4.10 Overlapping intervals are the wrong test [F46]

"Do the CIs overlap?" is a conservative approximation. Both retrievers are scored on the *same* impressions, so their shared per-impression noise cancels when you subtract — and a paired bootstrap on the *difference* is both the correct test and a much tighter one.

It changed two conclusions. The 384-vs-256 intervals above overlap heavily ([0.2025, 0.2613] against [0.2206, 0.2806]) and the overlap heuristic would have called it undecided; the paired test finds it significant, so we would have shipped the larger, worse vector. Conversely an earlier draft of this note claimed semantic "edges ahead" of lexical on EB-NeRD from the point estimates; paired, the difference is −0.0055 [−0.0277, +0.0155] — the sign reverses and nothing is established.

Every null result now reports how many impressions the two systems actually differ on, because that separates *no effect* from *no power*: the dedup ablation differed on **4 of 800** (unmeasurable by construction), while EB-NeRD semantic-vs-lexical differed on **470 of 800** — abundant signal, genuinely equal.

5 Anti-gaming (Q9)

Leakage. History is truncated to clicks strictly before the impression time. `test_no_leakage.py` asserts this on the built store *and injects a deliberate post-boundary click to confirm the checker fails* — a test that passes both before and after the boundary breaks proves nothing. *Honest limit:* verifiable only where per-click timestamps exist, i.e. EB-NeRD. MIND's history is untimestamped, so there the boundary rests on the authors' construction; the store records `history_boundary_verifiable: false` rather than implying otherwise.

Serving-unavailable features. EB-NeRD's training rows carry `read_time`, `scroll_percentage` and `next_read_time/next_` — all recorded *after* the click. Predicting a click from its own consequences is circular. The organisers settled the boundary: those four columns are exactly what is deleted from the test set. They are excluded in `clean.py` with a comment naming why. Our retrievers never touch them, so the honest with/without pair is the cold-start fallback flag, reported as a `features` column on every result row.

6 Where it breaks at 10×

- **The join breaks before the model does [F15].** EB-NeRD large is 52× small's impressions and its history parquet exceeds its behaviours file. The data layer gives first.
- **The prediction path needed a different algorithm, not a bigger machine [F32].** Full-corpus retrieval per impression ran at 162 impressions/s (∼4 h/file) because it discarded 99% of each result. Scoring the slate directly: **2,570/s, 16×.**
- **Parallelism was memory-bound, and the fix was representation — twice [F36,F64–F70].** A worker peaked at 15.1 GB, of which the index was only 1.8 GB; the rest was 116.8M click ids as Python `str`. Storing them as `array('i')` indices gave **9.25 GB and 45% faster**. Pushing the same idea further — holding histories as contiguous `int32` arrays read straight from parquet row groups, never as Python objects — cut the load from **160.8 s to 4.6 s** and ∼13 GB to **0.93 GB** [F64].
  That representation then enabled something the object version could not do. Every worker had been rebuilding the *same* immutable index, articles and histories, so N workers meant N copies (19.3 GB each, measured). Building them once in the parent and letting `fork`'s copy-on-write share them takes the true marginal cost of a worker — `Private_Dirty`, not `RSS` — to **0.38 GB, a 48× reduction** [F70]. This is only possible with arrays: CPython refcounting *writes* to an object header whenever a child touches it, dirtying the page and forcing a private copy,

whereas an array's data buffer is never touched by refcounting.

Configuration	workers	RAM	time
2 workers *(produced the submission)*	2	48 GB req. on 31 GB	∼49 min
Columnar histories [F64,F67]	3	13 GB	33 min
+ fork-shared state [F70]	**6**	**10 GB**	**22 min**

Every one of these produced a byte-identical submission (SHA-256 `066de46d...`, 13,336,711 lines) — the gate the work was held to, since a performance change that alters the output is a defect, not an optimisation. The 2-worker row is the configuration that thrashed: ∼140,000 blocks/s of swap I/O with all 32 cores 75–100% *idle*.

- **Swap does not slow the merge, it stops it [F38].** All 51 shards wrote at 6,426 lines/s, then the merge produced *zero* output for 20+ minutes with 24 GB swapped. The same merge with RAM free: **7 seconds**. Scoring streams and degrades gracefully; the merge hits a 13.3M-element set at random, where swap does not degrade — it stalls.
- **ANN: the defaults break at 10×, the method does not [F49–F53].** Two corrections got us here, and both were tuning rather than limits. First, an early benchmark on *uniform-random* vectors measured HNSW losing 11–66% of the exact answer; random vectors in 384 dimensions are near-orthogonal, the one input a proximity graph cannot exploit. On *clustered* vectors — the structure real embeddings have — the loss is 0.03–1.25%. Second, at 1M vectors a fixed `efSearch`=128 does collapse to **0.68** recall, because HNSW explores a fixed candidate count regardless of corpus size, so the *fraction* inspected shrinks as it grows.
  Sweeping both dials at 1M (exact baseline 22.8 ms/query):

M	`ef`	recall	ms/query	vs exact	graph
16	1024	0.9588	0.34	68×	0.13 GB
32	512	0.9793	0.34	67×	0.26 GB
64	**512**	**0.9947**	0.51	**45**×	0.52 GB
64	1024	0.9990	0.58	39×	0.52 GB

At *fixed latency* a denser graph always wins — more links means fewer dead ends, so the walk reaches the same recall exploring fewer candidates. **Shipped M=64 with `ef` derived from corpus size: 99.5% of the exact answer, 45× faster.** Index memory is not the binding constraint (0.52 GB of graph against 1.54 GB of vectors); build time is a 183 s one-off.

7 Leaderboards (Q5) and limitations

MIND submitted: **AUC 0.5568, MRR 0.2646, nDCG@5 0.2778, nDCG@10 0.3331**. EB-NeRD file generated and validated (13,336,711 lines, 100% valid permutations). *Screenshots overleaf.*

Calibration [F44]. The challenge paper (arXiv 2409.20483) publishes reference points for EB-NeRD: winners at 89.24 / 88.15 / 87.07 AUC, a *most-clicks* editorial baseline at **59.70**, read-time 59.49, in-view-rate 54.50, random 49.98. Our MIND fusion at 0.5934 sits essentially at the most-clicks baseline — an analogy across datasets, not a like-for-like, but a useful order of magnitude. All three winners used GBDT ensembles or transformers in multi-stage pipelines with engineered temporal features, i.e. a **re-ranker**: the 0.59→0.87 gap is a different class of system, not a better-tuned candidate generator. Notably their ablations independently identified *popularity* and *article timeliness* as the load-bearing signals — the same two levers this component found in [F16/F41] and [F25/F39].

Limitations, stated plainly. (i) The offline harness was under-powered: at n=800 it reported MIND AUC 0.4981 [0.4776, 0.5190] while the leaderboard scored **0.5568** — *outside* the interval [F34]. Now n=4,000+, but several reported differences still have overlapping CIs and are labelled as such. (ii) Abstracts are 75% of the lexical index and buy **+0.011 recall against a ±0.030 CI** — kept because Q2.1 mandates it, reported as the null result it is [F28]. (iii) MIND's test window is a single day, so one day's topical anomaly moves every MIND number. (iv) Reproduction is verified for both datasets by regenerating each submission and comparing SHA-256: MIND `ba275ef0...` in 2,282 s against a 2,288 s baseline, EB-NeRD `066de46d...`. *Caveat:* EB-NeRD's check ran on the optimised path (byte-identical, so the pipeline is deterministic), not on the exact 2-worker configuration that produced the upload — that configuration no longer fits this machine.

The methodological finding, which outranks the results. Offline and leaderboard disagreed **three times out of three, in both directions**: n=800 *understated* MIND (0.4981 vs 0.5568) [F34]; fusion was *not significant* offline yet gained **+0.0366** on 2.37M impressions [F42]; 256-d was significant offline at +0.0175 and moved the leaderboard **+0.0004** [F58]. The harness remains sound for what it was built for — leakage, slicing, the recency effect — but **no retriever or parameter choice here is settled until it has been submitted**, and this note labels each such claim rather than rounding it into a result.